

Project Falcon Quality Playbook

Standard Operating Procedure: who owns what in quality, and the principles behind our testing

QA (Quality Assurance)

Proactive and process-focused. Prevents defects by improving how we build.

QC (Quality Control)

Reactive and product-focused. Finds defects in what we have built.

1. Quality Control (QC) Responsibilities

Owner: QC Process Owner. Question answered: "Does the product we built work as specified?"

- **Design and execute test cases** against approved requirements, covering functional, integration, and edge-case scenarios.
- **Log and track defects** with clear reproduction steps, severity, and evidence, and verify each fix before closure.
- **Run regression tests** after every change or release candidate to confirm existing features still work.
- **Report product quality status** (pass/fail rates, open defects) so the team can make go/no-go release decisions.

2. Quality Assurance (QA) Responsibilities

Owners: QA testers. Question answered: "Are we building the product the right way?"

- **Set quality standards**, including definitions of done, coding and testing guidelines, and defect severity levels.
- **Review requirements and designs** before development starts to catch ambiguity, gaps, and untestable criteria.
- **Audit processes** such as code reviews, test documentation, and release checklists, and recommend improvements.
- **Analyze defect trends and root causes** to update the process so the same class of defect does not recur.

3. Our Guiding Principles

We ground our work in three of the seven fundamental testing principles.

- **Early testing.** Quality work starts at requirements, not after code is written. QA testers review every user story and acceptance criterion before it enters a sprint, because a defect found in requirements is far cheaper to fix than one found in production.
- **Defect clustering.** A small number of modules usually contain most of the defects. QC tracks defects by component each sprint, and the areas with the highest counts (new, complex, or frequently changed code) receive deeper testing and extra code review.
- **Pesticide paradox.** Running the same tests repeatedly stops finding new bugs. QC reviews and refreshes the regression suite every release, adding cases for new features and past escapes, retiring obsolete ones, and varying test data and approach.

Rule of thumb: If the work improves *how we build*, it is QA. If it checks *what we built*, it is QC. Quality is everyone's responsibility; these roles define who leads each activity.